

CT5102 Assignment 1 - Vectors (100 marks)

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Objective

The aim of this assignment is to generate synthetic temperature data for three months of the year, and then analyse the data. The tasks are based on the content of lecture 1 and lecture 2. This is an open-book assignment, however, access to Gen-AI tools is not permitted.

Create a list containing the temperature data (30 marks)

Here are the steps

- Set the seed to 100 (to ensure that you generate the same results as displayed here)
- Create a list with three elements, one for the observations for each month (January, February, and March)

For the data generation, assume the following (sample from a normal distribution using the R function `rnorm(n, mean, sd)`)

- For January (31 Days) assume a mean of **7** and a standard deviation of **5**
- For February (28 days), assume a **14** and a **7**

- For March (31 days), assume a **16** with a **3**
- The overall list length will be 3, and each month's data will be an atomic vector, and make sure that the atomic vectors have meaningful names (e.g Jan-1, ..., Jan 31, Feb-1, ... Feb-28, Mar-1, ..., Mar-31)

Here is information on the list data structure you should have created.

```
str(temp_list)
```

List of 3

```
$ JAN: Named num [1:31] 4.5 7.7 6.6 11.4 7.6 8.6 4.1 10.6 2.9 5.2 ...
..- attr(*, "names")= chr [1:31] "Jan-1" "Jan-2" "Jan-3" "Jan-4" ...
$ FEB: Named num [1:28] 26.3 13 13.2 9.2 12.4 15.3 16.9 21.5 20.8 13.3 ...
..- attr(*, "names")= chr [1:28] "Feb-1" "Feb-2" "Feb-3" "Feb-4" ...
$ MAR: Named num [1:31] 13.5 15.2 15.8 14.9 23.7 16.4 13.9 17.9 16.6 15.8 ...
..- attr(*, "names")= chr [1:31] "Mar-1" "Mar-2" "Mar-3" "Mar-4" ...
```

```
names(temp_list)
```

```
[1] "JAN" "FEB" "MAR"
```

```
names(temp_list$JAN)
```

```
[1] "Jan-1" "Jan-2" "Jan-3" "Jan-4" "Jan-5" "Jan-6" "Jan-7" "Jan-8"
[9] "Jan-9" "Jan-10" "Jan-11" "Jan-12" "Jan-13" "Jan-14" "Jan-15" "Jan-16"
[17] "Jan-17" "Jan-18" "Jan-19" "Jan-20" "Jan-21" "Jan-22" "Jan-23" "Jan-24"
[25] "Jan-25" "Jan-26" "Jan-27" "Jan-28" "Jan-29" "Jan-30" "Jan-31"
```

```
names(temp_list$FEB)
```

```
[1] "Feb-1" "Feb-2" "Feb-3" "Feb-4" "Feb-5" "Feb-6" "Feb-7" "Feb-8"
[9] "Feb-9" "Feb-10" "Feb-11" "Feb-12" "Feb-13" "Feb-14" "Feb-15" "Feb-16"
[17] "Feb-17" "Feb-18" "Feb-19" "Feb-20" "Feb-21" "Feb-22" "Feb-23" "Feb-24"
[25] "Feb-25" "Feb-26" "Feb-27" "Feb-28"
```

```
names(temp_list$MAR)
```

```
[1] "Mar-1" "Mar-2" "Mar-3" "Mar-4" "Mar-5" "Mar-6" "Mar-7" "Mar-8"
[9] "Mar-9" "Mar-10" "Mar-11" "Mar-12" "Mar-13" "Mar-14" "Mar-15" "Mar-16"
[17] "Mar-17" "Mar-18" "Mar-19" "Mar-20" "Mar-21" "Mar-22" "Mar-23" "Mar-24"
[25] "Mar-25" "Mar-26" "Mar-27" "Mar-28" "Mar-29" "Mar-30" "Mar-31"
```

Convert the list to an atomic vector (10 marks)

Using an appropriate R function, convert `temp_list` to an atomic vectors `temp_atvec`. The function `unname()` can be used to remove the three names of the list.

```
temp_atvec
```

Jan-1	Jan-2	Jan-3	Jan-4	Jan-5	Jan-6	Jan-7	Jan-8	Jan-9	Jan-10	Jan-11
4.5	7.7	6.6	11.4	7.6	8.6	4.1	10.6	2.9	5.2	7.4
Jan-12	Jan-13	Jan-14	Jan-15	Jan-16	Jan-17	Jan-18	Jan-19	Jan-20	Jan-21	Jan-22
7.5	6.0	10.7	7.6	6.9	5.1	9.6	2.4	18.6	4.8	10.8
Jan-23	Jan-24	Jan-25	Jan-26	Jan-27	Jan-28	Jan-29	Jan-30	Jan-31	Feb-1	Feb-2
8.3	10.9	2.9	4.8	3.4	8.2	1.2	8.2	6.5	26.3	13.0
Feb-3	Feb-4	Feb-5	Feb-6	Feb-7	Feb-8	Feb-9	Feb-10	Feb-11	Feb-12	Feb-13
13.2	9.2	12.4	15.3	16.9	21.5	20.8	13.3	23.8	1.6	18.4
Feb-14	Feb-15	Feb-16	Feb-17	Feb-18	Feb-19	Feb-20	Feb-21	Feb-22	Feb-23	Feb-24
10.3	23.3	11.5	23.2	14.3	0.8	10.9	1.8	15.3	27.3	-1.9
Feb-25	Feb-26	Feb-27	Feb-28	Mar-1	Mar-2	Mar-3	Mar-4	Mar-5	Mar-6	Mar-7
20.9	4.2	26.8	23.7	13.5	15.2	15.8	14.9	23.7	16.4	13.9
Mar-8	Mar-9	Mar-10	Mar-11	Mar-12	Mar-13	Mar-14	Mar-15	Mar-16	Mar-17	Mar-18
17.9	16.6	15.8	15.7	17.3	12.8	12.5	20.9	9.8	16.0	12.7
Mar-19	Mar-20	Mar-21	Mar-22	Mar-23	Mar-24	Mar-25	Mar-26	Mar-27	Mar-28	Mar-29
16.8	19.0	9.8	18.7	15.9	12.0	10.2	18.1	15.5	16.6	18.5
Mar-30	Mar-31									
21.2	15.7									

```
length(temp_atvec)
```

```
[1] 90
```

Display summary statistics on the overall data set (10 marks)

```
s_data
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-1.90	7.60	12.60	12.54	16.75	27.30

Calculate the weather profiles for the three months (30 marks)

Each observation can be divided into a number of categories

- A cold day (less than 8 degrees)
- A mild day (between 9 and 19 degrees)
- A warm day (greater than 19 degrees)

Create a new atomic vector that contains this information, and it should look like this.

```
temp_categories
```

```
Jan-1 Jan-2 Jan-3 Jan-4 Jan-5 Jan-6 Jan-7 Jan-8 Jan-9 Jan-10 Jan-11
"Cold" "Cold" "Cold" "Mild" "Cold" "Mild" "Cold" "Mild" "Cold" "Cold" "Cold"
Jan-12 Jan-13 Jan-14 Jan-15 Jan-16 Jan-17 Jan-18 Jan-19 Jan-20 Jan-21 Jan-22
"Cold" "Cold" "Mild" "Cold" "Cold" "Cold" "Mild" "Cold" "Mild" "Cold" "Mild"
Jan-23 Jan-24 Jan-25 Jan-26 Jan-27 Jan-28 Jan-29 Jan-30 Jan-31 Feb-1 Feb-2
"Mild" "Mild" "Cold" "Cold" "Cold" "Mild" "Cold" "Mild" "Cold" "Warm" "Mild"
Feb-3 Feb-4 Feb-5 Feb-6 Feb-7 Feb-8 Feb-9 Feb-10 Feb-11 Feb-12 Feb-13
"Mild" "Mild" "Mild" "Mild" "Mild" "Warm" "Warm" "Mild" "Warm" "Cold" "Mild"
Feb-14 Feb-15 Feb-16 Feb-17 Feb-18 Feb-19 Feb-20 Feb-21 Feb-22 Feb-23 Feb-24
"Mild" "Warm" "Mild" "Warm" "Mild" "Cold" "Mild" "Cold" "Mild" "Warm" "Cold"
Feb-25 Feb-26 Feb-27 Feb-28 Mar-1 Mar-2 Mar-3 Mar-4 Mar-5 Mar-6 Mar-7
"Warm" "Cold" "Warm" "Warm" "Mild" "Mild" "Mild" "Mild" "Warm" "Mild" "Mild"
Mar-8 Mar-9 Mar-10 Mar-11 Mar-12 Mar-13 Mar-14 Mar-15 Mar-16 Mar-17 Mar-18
"Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Warm" "Mild" "Mild" "Mild"
Mar-19 Mar-20 Mar-21 Mar-22 Mar-23 Mar-24 Mar-25 Mar-26 Mar-27 Mar-28 Mar-29
"Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild" "Mild"
Mar-30 Mar-31
"Warm" "Mild"
```

Now, using the appropriate R function, generate a frequency count of the different categories.

```
freq_count
```

```
temp_categories
Cold Mild Warm
  25   52   13
```

Querying specific days (20 marks)

Display the following information

- The coldest day
- The warmest day
- The days when the weather was cold (hint make use of `which()`)
- The days when the weather was mild or warm (hint make use of `which()` and logical operator)

`coldest_day`

Feb-24
-1.9

`warmest_day`

Feb-23
27.3

`days_cold`

Jan-1	Jan-2	Jan-3	Jan-5	Jan-7	Jan-9	Jan-10	Jan-11	Jan-12	Jan-13	Jan-15
4.5	7.7	6.6	7.6	4.1	2.9	5.2	7.4	7.5	6.0	7.6
Jan-16	Jan-17	Jan-19	Jan-21	Jan-25	Jan-26	Jan-27	Jan-29	Jan-31	Feb-12	Feb-19
6.9	5.1	2.4	4.8	2.9	4.8	3.4	1.2	6.5	1.6	0.8
Feb-21	Feb-24	Feb-26								
1.8	-1.9	4.2								

`days_mild_or_warm`

Jan-4	Jan-6	Jan-8	Jan-14	Jan-18	Jan-20	Jan-22	Jan-23	Jan-24	Jan-28	Jan-30
11.4	8.6	10.6	10.7	9.6	18.6	10.8	8.3	10.9	8.2	8.2
Feb-1	Feb-2	Feb-3	Feb-4	Feb-5	Feb-6	Feb-7	Feb-8	Feb-9	Feb-10	Feb-11
26.3	13.0	13.2	9.2	12.4	15.3	16.9	21.5	20.8	13.3	23.8
Feb-13	Feb-14	Feb-15	Feb-16	Feb-17	Feb-18	Feb-20	Feb-22	Feb-23	Feb-25	Feb-27
18.4	10.3	23.3	11.5	23.2	14.3	10.9	15.3	27.3	20.9	26.8
Feb-28	Mar-1	Mar-2	Mar-3	Mar-4	Mar-5	Mar-6	Mar-7	Mar-8	Mar-9	Mar-10
23.7	13.5	15.2	15.8	14.9	23.7	16.4	13.9	17.9	16.6	15.8
Mar-11	Mar-12	Mar-13	Mar-14	Mar-15	Mar-16	Mar-17	Mar-18	Mar-19	Mar-20	Mar-21

15.7	17.3	12.8	12.5	20.9	9.8	16.0	12.7	16.8	19.0	9.8
Mar-22	Mar-23	Mar-24	Mar-25	Mar-26	Mar-27	Mar-28	Mar-29	Mar-30	Mar-31	
18.7	15.9	12.0	10.2	18.1	15.5	16.6	18.5	21.2	15.7	